

Note 1 Solutions

Question 1: NAND Gate Constructions

Recall that the NAND operation is defined as:

$$\text{NAND}(x, y) = \neg(x \wedge y)$$

Part (a): NOT using NAND

Claim: $\neg x = \text{NAND}(x, x)$.

Proof:

$$\text{NAND}(x, x) = \neg(x \wedge x) = \neg x$$

since $x \wedge x = x$ by idempotence. □

Part (b): AND using NAND

Claim: $x \wedge y = \text{NAND}(\text{NAND}(x, y), \text{NAND}(x, y))$.

Proof:

We want to represent $x \wedge y$ using only NAND gates. Observe:

$$x \wedge y = \neg(\neg(x \wedge y)) = \neg(\text{NAND}(x, y))$$

From part (a), negation can be expressed as $\neg z = \text{NAND}(z, z)$. Substituting $z = \text{NAND}(x, y)$:

$$x \wedge y = \text{NAND}(\text{NAND}(x, y), \text{NAND}(x, y))$$

□

Part (c): OR using NAND

Claim: $x \vee y = \text{NAND}(\text{NAND}(x, x), \text{NAND}(y, y))$.

Proof:

By De Morgan's Law:

$$x \vee y = \neg(\neg x \wedge \neg y)$$

From part (a), $\neg x = \text{NAND}(x, x)$ and $\neg y = \text{NAND}(y, y)$. Therefore:

$$x \vee y = \neg(\text{NAND}(x, x) \wedge \text{NAND}(y, y)) = \text{NAND}(\text{NAND}(x, x), \text{NAND}(y, y))$$

□

Question 2: Winning Strategies in Two-Player Games

Consider a two-player game between Toby and Fritz. The game consists of two turns: Toby moves first, then Fritz, and exactly one player is declared the winner (no ties).

Part (a): One Player Must Have a Winning Strategy

Claim: Either Toby or Fritz has a winning strategy.

Proof:

Define the proposition P :

$$P = \text{“}\exists \text{ a move by Toby such that } \forall \text{ moves by Fritz, Toby wins.”}$$

By the **Law of Excluded Middle**, either P is true or $\neg P$ is true.

Case 1: P is true.

Then Toby has a move that guarantees a win regardless of how Fritz responds. This is precisely a winning strategy for Toby.

Case 2: $\neg P$ is true.

The negation of P is:

$$\neg P = \text{“}\forall \text{ moves by Toby, } \exists \text{ a move by Fritz such that Toby does not win.”}$$

Since there are no ties, “Toby does not win” is equivalent to “Fritz wins.” Therefore $\neg P$ states:

$$\forall \text{ moves by Toby, } \exists \text{ a move by Fritz such that Fritz wins.}$$

This means Fritz can always respond to any Toby move with a winning move — i.e., Fritz has a winning strategy.

In both cases, one of the two players has a winning strategy. □

Part (b): Which Property Enables the Argument

The property that allowed us to apply the Law of Excluded Middle is that **there are no ties**: exactly one player wins every game. This ensures that the proposition “Toby wins” is a well-defined Boolean — it is either true or false, with no third possibility. Without this binary outcome, the Law of Excluded Middle could not be cleanly applied, as a tie would represent a case where neither “Toby wins” nor “Fritz wins” is true.